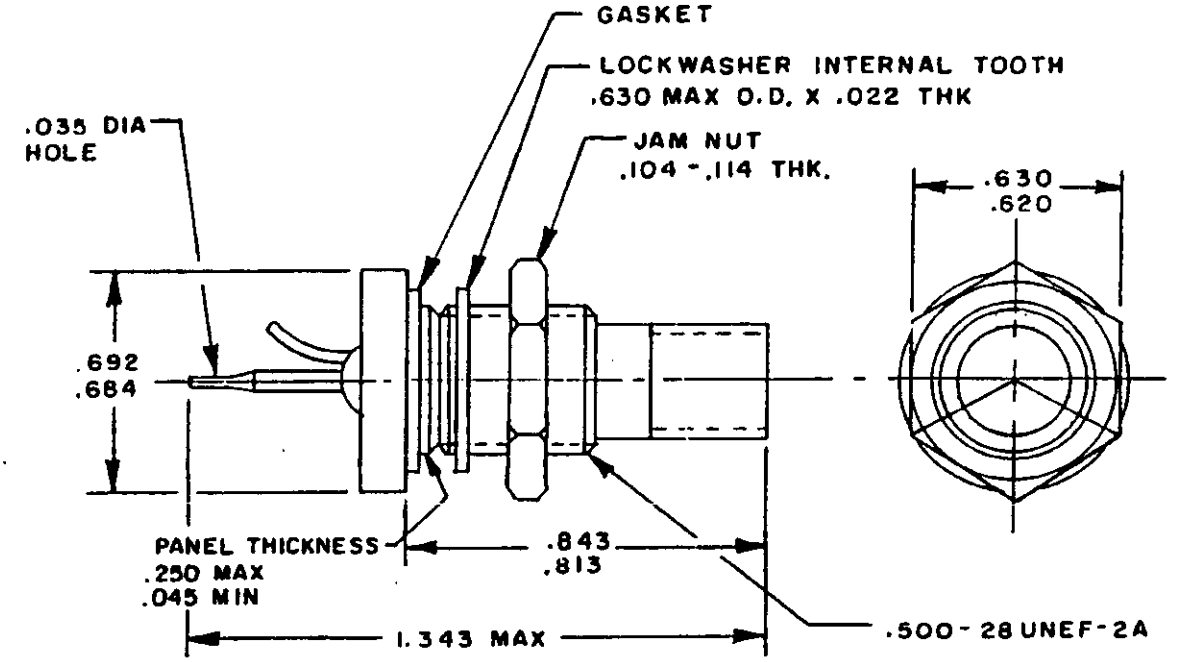
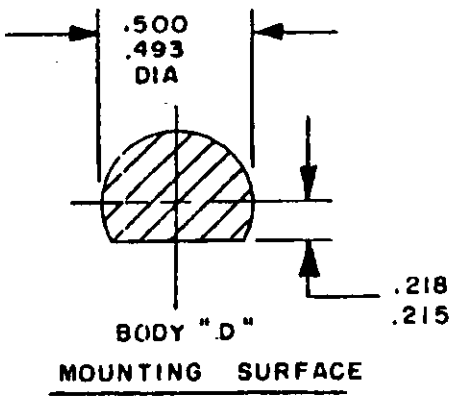
MILITARY SPECIFICATION SHEET

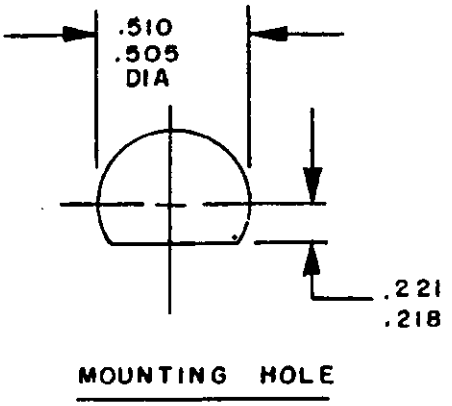
CONNECTORS, TRIAXIAL, RADIO FREQUENCY, (SERIES TRT (UNCABLED) RECEPTACLE, SOCKET CONTACT, JAMNUT MOUNTED, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connector described herein shall consist of this specification and the latest issue of MIL-C-49142.







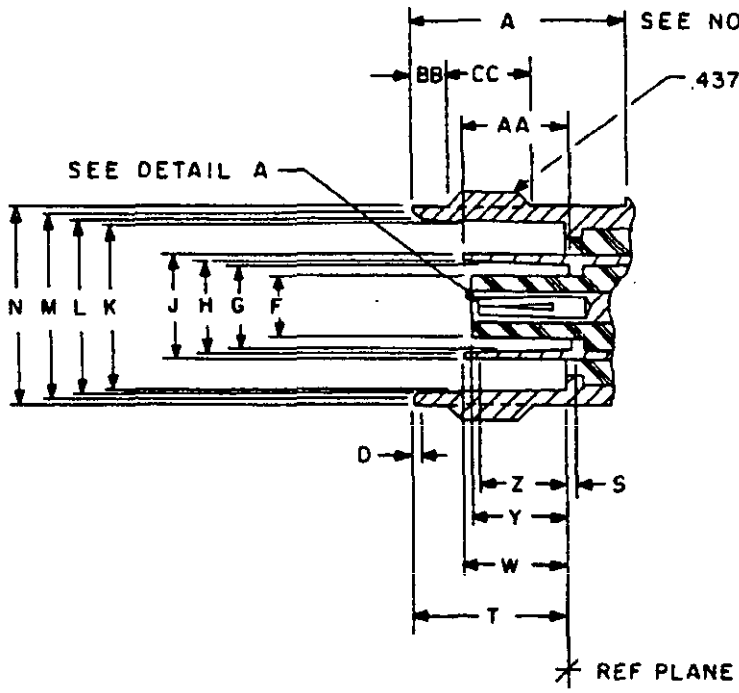
Inches	mm
.022	.56
.035	.88
.045	1.14
.104	2.64
.114	2.50
.215	5.46
.218	5.54
.250	6.35
.221	5.61
.493	12.52
.500	12.70
.505	12.71
.510	12.95
.620	15.75
.630	16.00
.684	17.37
.692	17.58
.813	20.65
.843	21.41
1.343	34.11

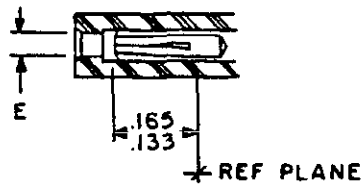
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Wrench flats are to accommodate standard wrench openings per FED-STD-H28, Appendix 10.
4. All undimensioned pictorial representations are for reference purposes only.
5. Unless otherwise specified all tolerances are ±.005 inch.

FIGURE 1. General configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

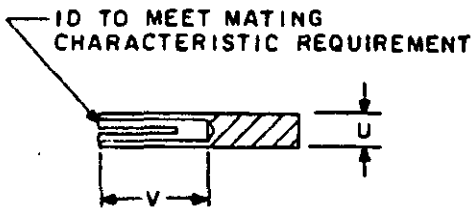
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ALTERNATE CLOSED ENTRY INSULATOR

DETAIL A



CONTACT

Ltr	Inches		Millimeters	
	Min	Max	Min	Max
A	.414	---	10.52	---
D	.015	.030	0.38	0.76
E	.040	.043	0.99	1.09
F	.117	.122	2.97	3.10
G	.169	.171	4.29	4.34
H	.178	.182	4.52	4.62
J	.195	.199	4.95	5.05
K	.319	.321	8.10	8.15
L	.327	.333	8.31	8.46
M	.346	.356	8.79	9.04
N	.378	.382	9.50	9.70
S	.001	---	.03	---
T	.327	.335	8.31	8.51
U	.062	.064	1.57	1.63
V	.213	---	5.41	---
W	.187	.213	4.75	5.41
Y	---	.213	---	5.41
Z	.165	.263	4.19	5.16
AA	.206	.213	5.23	5.41
BB	.068	.088	1.73	2.24
CC	.187	---	4.75	---

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are tabulated.
4. Chamfer or radius.
5. Clearance for mating connector coupling nut.

FIGURE 2. Interface series TRT, socket contact.

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ENGINEERING DATA:

Nominal impedance: Non constant.

Frequency range: 0 to 500 MHz minimum.

Voltage rating: 400 V rms, maximum working voltage at sea level. 100 V rms, maximum working voltage at 70,000 feet.

Temperature range: -65°C to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figures 1 and 2.

Force to engage and disengage:
Longitudinal force - Not applicable.
Torque - 2.5 inch-pounds maximum.

Coupling proof torque: Not applicable.

Inspection conditions: Coupling torque - 4-6 inch-pounds.

Mating characteristics: See figure 2 for dimensions.
Center contact (socket):
Oversize test pin - 0.40 inch diameter, minimum (nonclosed entry contacts only).
Insertion depth - .125 inch minimum.
Number of insertions - 1.
Insertion force test:
Steel test pin finish - 16 microinches.
Insertion force - 2 pounds maximum.
Steel test pin diameter - .039 minimum.
Withdrawal force test:
Steel test pin diameter - .037 maximum.
Withdrawal force - 2 ounces minimum.
Test pin finish - 16 microinches.

Permeability: Applicable.

Hermetic seal: Not applicable.

Leakage: Not applicable.

Insulation resistance: 5000 megohms.

Conductor retention: 6 pounds minimum, axial force.

Dielectric withstanding voltage:. At sea level, 1200 V rms, between center conductor and intermediate conductor. 500 V rms, between intermediate conductor and outer conductor.

Salt spray (corrosion)): Applicable.

Vibration: Applicable.

Shock: Applicable.

Thermal shock: Applicable.

Moisture resistance: Applicable.

Conductor resistance: In milliohms, maximum.

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	Initial	After environment
Center conductor	2.0	2.5
Intermediate conductor	0.5	0.6
Outer conductor	0.2	0.3

Corona level:
Altitude - 70,000 feet.
Voltage - 200 V rms minimum.

RF high potential withstanding voltage:
800 V rms, between center conductor and intermediate conductor.
200 V rms, between intermediate conductor and outer conductor at 5 to 7.5 MHz.

Leakage current: Not applicable.

Cable retention force (for cable .200 to .325 outside dimension): 40 pounds, minimum.

Coupling mechanism retention force: Not applicable.

Risetime degradation: To be added.

Connector durability: 500 cycles minimum at 12 cycles per minute maximum.

Part number M49142/10- (dash number from table I.)

TABLE I. Dash number, superseded part number.

Dash number	Superseded part number
-0001	---
-0002 (Hermetic)	---

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT
Navy - AS, MC
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 5935-3662-02)

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